ADITYA UNIVERSITY

SCHOOL OF ENGINEERING

Department of Computer Science and Engineering

CERTIFICATE

This is to certify that the Cornerstone project work entitled *"* **ONLINE STUDENT RESULT MANAGEMENT SYSTEM** *"* is submitted by

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in partial fulfillment of the requirements for the award of the B.Tech. degree in Computer Science and Engineering for the academic year 2025-2026.

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DECLARATION

We hereby declare that the Cornerstone project entitled **"ONLINE STUDENT RESULT MANAGEMENT SYSTEM"** is original and is submitted to **ADITYA UNIVERSITY**, Surampalem, in partial fulfillment of the requirements for the award of the **B.Tech. degree in Computer Science and Engineering**.

We further declare that this project work has not been submitted, either in full or in part, for the award of any degree in this or any other institution. We also confirm that this work complies with the Institutional Academic Integrity and AI Usage Policy.

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ACKNOWLEDGEMENT

It is with immense pleasure that we express our sincere gratitude to our Cornerstone Project Guide, **Dr. K Sreenu, Assistant Professor**, who has guided us throughout the project. Her valuable support, encouragement, and guidance have greatly contributed to the successful completion of this work.

We would like to thank our Deputy Heads of the Department, **Dr. Chandra Sekhar Kolli, Associate Professor and Dr. Hari Jyothula , Associate Professor**, Department of CSE, for their support and valuable suggestions.

Our deepest thanks to our Head of the Department **Dr. T. Sudha Rani, Assistant Professor,** for her inspiration and for providing all the necessary facilities and resources required for this project.

We also extend our sincere thanks to **Dr. M.V. Rajesh, Associate Dean, School of Engineering** for their encouragement and support during the course of our project.

We would like to express our sincere gratitude to **Dr. G. Suresh, Registrar, Dr. A Ramesh, Pro Vice-Chancellor (Engineering & Sciences), Dr. S. Rama S r e e , Pro Vice-Chancellor (Academics), D r . M.B. Srinivas, V i c e -Chancellor, Dr. M. Sreenivasa Reddy, Deputy Pro Chancellor and Management, Aditya University** for providing excellent infrastructure and state-of-the-art laboratories.

Finally, we would like to thank the **faculty members, lab technicians, non-teaching staff, and our friends** who have directly or indirectly supported us in completing this Cornerstone project successfully.

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ABSTRACT

The **Online Student Result Management System** is a Java-based application developed to manage student records and academic results efficiently. The system provides functionalities such as adding student details, managing subject-wise marks, calculating grades and credits automatically, and displaying results in a structured format.

The application is developed using Core Java and JDBC for database connectivity, with MySQL as the backend database. It ensures data accuracy, reduces manual effort, and improves accessibility of student records.

The system supports two roles: Admin and Student. Admin can manage student and result data, while students can securely log in to view their results. This project provides a simple, reliable, and efficient solution for managing academic records digitally.

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ACRONYMS

- **SRMS : Student Result Management System**
- **JDBC : Java Database Connectivity**
- **SQL : Structured Query Language**
- **DBMS : Database Management System**
- **GUI : Console Based**

1. INTRODUCTION

1.1 Brief Information about the Project

The Student Result Management System is a console-based application designed to manage student details and academic results efficiently. It provides a centralized system to store, update, and retrieve student information and subject-wise marks.

The system is developed using Core Java and JDBC, with MySQL as the backend database. It allows administrators to manage records and students to view their results securely.

1.2 Motivation and Contribution of Project

Manual result management systems are time-consuming and prone to errors.

This project was developed to:

- Reduce manual work
- Avoid calculation errors
- Provide quick result access
- Maintain centralized data

1.3 Objectives of the Project

The main objectives of the Online Student Result Management System are:

- To develop a system to manage student and result data
- To automate grade and credit calculation
- To provide secure login for students
- To enable easy retrieval of results

1.4 Scope of the project

The system can be used in colleges and institutions for managing student academic records. It supports operations like adding students, managing results, and viewing performance reports.

2. SYSTEM ANALYSIS

2.1 Functional Requirements

Functional requirements define the essential operations and functionalities that the **Student Result Management System (SRMS)** must perform to manage student data and academic results efficiently. These requirements include managing student records, maintaining subject-wise results, performing result calculations, and providing secure login access for both administrators and students.

- **Add Student Details**

  The system allows the administrator to add new student records into the database. Details such as roll number, name, department, semester, section, and password are entered and stored. Validation is applied to avoid duplicate entries and ensure data accuracy.

- **Update Student Records**

  The system provides functionality to update existing student details. The administrator can modify fields such as name, department, semester, and section whenever changes are required.

- **Delete Student Information**

  The administrator can remove a student record from the database using the roll number. This feature helps in maintaining updated records by deleting inactive or incorrect entries.

- **Search Student**

  The system allows searching for a specific student using the roll number. It retrieves and displays relevant student details quickly from the database.

- **Add Result**

  The administrator can add subject-wise results for students. The system stores course code, course name, and marks. It automatically calculates grade and credits based on marks.

- **Update Result**

  The system allows updating the marks of a specific subject for a student. When marks are updated, the system automatically recalculates the grade and credits.

- **Delete Result**

  The administrator can delete a specific subject result using roll number and course code. This helps in correcting errors or removing unwanted records.

- **View Results**

  The system displays all subject results of a student in a structured format. It shows marks, grades, credits, total marks, and average. It also indicates whether the student has passed or failed.

- **Login System**

  The system provides a login feature for both admin and students. Admin can access full functionalities, while students can log in using their roll number and password to view their results securely

2.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards that the **Student Result Management System** must satisfy.

- **Performance**

  The system is designed to respond quickly to user operations such as searching, adding, and updating records. Efficient database queries ensure minimal delay.

- **Security**

  The system ensures secure access by validating user credentials. Only authorized users can access sensitive data, preventing unauthorized usage.

- **Usability**

  The application provides a simple and user-friendly interface, making it easy for users to perform operations without technical knowledge.

- **Reliability**

  The system ensures proper database connectivity and error handling. It maintains data consistency and prevents data loss during operations.

- **Scalability**

  The system should handle increasing data and users without affecting performance.

2.3 Requirements Specification

This section specifies the hardware and software needed to run the system.

2.3.1 Minimum Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB (minimum)
- Hard Disk: 20 GB free space
- Monitor: Standard display

2.3.2 Software Requirements

- **Operating System:** Windows 7/10/11
- **Programming Language:** Java
- **IDE:** Intellij IDEA
- **Database:** MySQL Database
- **Connectivity:** JDBC

3. TECHNOLOGY DESCRIPTION

3.1 Programming Language

3.1.1 Introduction to Java

Java is a high-level, object-oriented programming language developed by Sun Microsystems. It is widely used for developing desktop, web, and enterprise applications due to its simplicity, portability, and robustness.

In the Student Result Management System, Java is used to implement the core functionality of the application. It handles operations such as adding, updating, deleting, and searching student records, as well as managing subject-wise results. Java supports object-oriented programming concepts such as classes, objects, inheritance, and encapsulation, which help in organizing the code efficiently.

One of the major advantages of Java is its platform independence. Programs written in Java can run on any system that has a Java Virtual Machine (JVM), making it highly portable. Java also provides strong exception handling mechanisms, which improve the reliability of the application by handling runtime errors effectively.

Overall, Java plays a crucial role in building a secure, scalable, and efficient system for managing student results.

3.2 Framework / Libraries

3.2.1 Introduction to Console Based

The Student Result Management System uses a console-based user interface instead of a graphical interface. The system interacts with users through text-based input and output using the command line.

Users can enter data and select options using the keyboard, and the system displays the results in a structured format on the console screen. The interface is simple, efficient, and easy to use without requiring any graphical components.

The system provides different options through menus, such as:

- Login (Admin / Student)
- Add Student
- Update Student
- Delete Student
- Add Result
- View Results

- Search Student

The console-based approach reduces system complexity and makes the application lightweight and easy to execute on any system. It is especially useful for learning core programming concepts and database integration without focusing on graphical design.

Although it does not include visual components like buttons or forms, the system ensures clear interaction through well-structured menus and prompts.

3.3 Database Technology

The Student Result Management System uses a relational database for storing and managing student and result information in a structured manner. The database ensures efficient data storage, retrieval, and manipulation, which is essential for maintaining accuracy and consistency in the system.

In this project, MySQL is used as the backend database, and JDBC (Java Database Connectivity) is used to connect the Java application to the database. JDBC acts as a bridge between the application and the database, allowing execution of SQL queries such as INSERT, UPDATE, DELETE, and SELECT.

The use of a relational database helps in organizing data into tables such as students and results, reducing redundancy and improving data integrity.

3.3.1 JDBC (Java Database Connectivity)

JDBC (Java Database Connectivity) is an API used to connect Java applications with databases. It provides methods and classes that allow developers to execute SQL queries and interact with the database.

In this project, JDBC is used to establish a connection between the Java application and the MySQL database. It acts as a bridge that enables communication between the application and the database.

The main functionalities of JDBC in this system include:

- Establishing a connection to the database
- Executing SQL queries such as INSERT, UPDATE, DELETE, and SELECT
- Retrieving data from the database
- Closing the database connection after operations

JDBC ensures efficient data handling and maintains consistency between the application and the database. It also supports the use of PreparedStatement, which improves security by preventing SQL injection.

3.3.2 MySQL Database

MySQL is a widely used relational database management system that stores data in a structured format using tables. It is known for its speed, reliability, and ease of use.

In the Student Result Management System, MySQL is used as the backend database to store student and result information. The database consists of tables such as:

Students Table – stores student details like roll number, name, department, semester, section, and password

Results Table – stores subject-wise marks, grades, and credits

MySQL supports Structured Query Language (SQL), which is used to perform operations such as inserting, updating, deleting, and retrieving data. It also ensures data integrity through constraints like primary keys and foreign keys.

The use of MySQL helps in organizing data efficiently and allows quick access to student records and results.

3.4 Tools & IDEs

3.4.1 IntellIJ IDEA

IntelliJ IDEA is a powerful Integrated Development Environment (IDE) developed by JetBrains, widely used for Java application development. It provides an efficient platform for writing, compiling, and debugging code with advanced features that enhance developer productivity.

In the Student Result Management System, IntelliJ IDEA is used to develop and manage the entire application. It offers a user-friendly interface along with intelligent coding assistance, making development faster and more efficient.

The key features of IntelliJ IDEA include:

- Syntax highlighting for improved code readability
- Smart code completion to speed up development
- Built-in debugging tools to identify and fix errors
- Automatic error detection and suggestions
- Integrated support for version control systems
- Easy project management and navigation

IntelliJ IDEA also provides strong support for Java and JDBC integration, which helps in seamless database connectivity and execution of SQL operations within the project.

Overall, IntelliJ IDEA simplifies the development process and improves code quality, making it an ideal tool for building the Student Result Management System.

4. DATABASE DESIGN

4.1 Database Tables

The Student Result Management System uses a relational database to store student and academic result information in a structured manner. The database consists of two main tables: **Students** and **Results**.

- The **Students table** stores details of students such as roll number, name, department, semester, section, and password.
- The **Results table** stores subject-wise marks, grades, and credits of students. It is linked to the Students table using the roll number.

This separation of data helps in reducing redundancy and improves data organization, making it easier to manage and retrieve information efficiently.

4.2 Table Structure

4.2.1 Students Table

Field Name	Data Type	Description	Constraint
roll_no	VARCHAR(20)	Student Roll No	Primary Key
name	VARCHAR(100)	Student Name	Not NULL
dept	VARCHAR(50)	Department	-
sem	INT	Semester	-
section	VARCHAR(10)	Section	-
password	VARCHAR(50)	Login Password	-

Table 1: Students Relational Database Table

4.2.2 Results Table:

Field Name	Data Type	Description	Constraint
id	INT	Unique ID	Primary Key
roll_no	VARCHAR(20)	Student Roll No	Foreign Key
Course_code	VARCHAR(20)	Course Role	-
Course_name	VARCHAR(100)	Course Name	-
marks	INT	Marks Obtained	-
grade	VARCHAR(5)	Grade	-
credits	DOUBLE	Credits	-

Table 2: Results Relational Database Table

4.3 Keys and Constraints

Keys and constraints are used to maintain data integrity and ensure accurate data storage.

- **Primary Keys:**
  - Emp_ID in Employee table
  - Dept_ID in Department table
- **Foreign Key:**
  - Dept_ID in the Employee table references Dept_ID in the Department table
- **NOT NULL Constraint:**
  - Important fields such as employee ID, name, and department ID cannot be left empty
- **Data Type Constraints:**
  - Each field accepts only valid data types to ensure correctness

These constraints help in maintaining consistency and avoiding duplicate or invalid data.

4.4 Table Relationships

The Employee and Department tables are related through the Dept_ID field.

- Each employee belongs to one department
- One department can have multiple employees

This represents a one-to-many relationship between Department and Employee tables.

5. SYSTEM DESIGN

5.1 System Architecture

The **Student Result Management System** follows a three-layer architecture to ensure proper organization and separation of functionalities.

1. **Presentation Layer (User Interface)**
   - Provides console-based interface
   - Allows users to interact through menus
   - Accepts input from users
   - Displays output in structured format
2. **Application Layer (Business Logic)**
   - Processes user inputs
   - Performs operations like add, update, delete, search
   - Contains classes like:
     - StudentDAO
     - ResultDAO
3. **Data Layer (Database)**
   - Stores student and result data
   - Uses MySQL database
   - JDBC used for connectivity
   - Executes SQL queries

5.2 Modules Description

The system is divided into the following modules:

1. Student Management Module

- Add new student details
- Update student information
- Delete student records
- Search student by roll number
- View all students

2. Result Module

- Add subject-wise results
- Update marks
- Auto-calculate grade and credits
- Delete results

- View results
- Identify failures

3. Login Module

- Provides authentication
- Admin login (username & password)
- Student login (roll number & password)
- Restricts access based on role

4. Database Connectivity Module

- Establishes connection using JDBC
- Executes SQL queries
- Retrieves and stores data

5. Integration Module

- Connects all modules
- Ensures smooth data flow
- Handles system execution
- Supports testing

6. SYSTEM IMPLEMENTATION

Login Screen:

Displays the login interface where the user enters username and password.

```
===== LOGIN =====
1. Admin
2. Student
Enter choice:
```

Figure 1: Login Screen of Student Result Management System

Main Menu Screen:

Provides options such as Add, Update, Delete, Search, and View student.

```
===== ADMIN MENU =====
1. Add Student
2. View Students
3. Search Student
4. Update Student
5. Delete Student
6. Add Result
7. Update Result
8. Delete Result
9. View Results
10. Search Result by Roll
11. Failures by Section
12. Failures by Subject
13. Exit
Choice:
```

Figure 2: Main Menu Screen of Student Result Management System

Add Student Screen:

Shows the form used to enter new student details.

```
Choice: 1
Roll No: 24B11CS126
Name: G.PRASANNA
Dept: CSE
Sem: 4
Section: 3
Password: PRASANNA@2007
✅ Student Added
```

Figure 3: Student Addition Screen of Student Result Management System

Update Student Screen:

Displays the interface for modifying existing student details.

```
Choice: 4
Roll No: 24B11CS126
New Name: PRASANNA
New Dept: CIVIL
New Sem: 4
New Section: 9
✅ Student Updated
```

Figure 4: Student Updation Screen of Student Result Management System

View Student Screen

Displays all Student records in a table format.

```
Choice: 2
24B11CS126 | PRASANNA | CIVIL | 4 | 9
24B11CS383 | R.ANUHYA | CSE | 4 | 5
24B11CS508 | Varun | CSE | 4 | 5
```

Figure 5: Student Information Screen of Student Result Management System

Delete Student Screen:

Shows the interface used to delete an Student record.

```
Choice: 5
Enter Roll No: 24B11CS126
✅ Student Deleted
```

Figure 6: Student Deletion Screen of Student Result Management System

Add Student Result Screen:

Shows the form used to enter student Result details.

```
Choice: 6
Roll No: 24B11CS508
Course Code: 241CS01
Course Name: ADSA
Marks: 97
✅ Subject Added
```

Figure 7: Student Result Addition Screen of Student Result Management System

Update Student Result Screen:

Displays the interface for modifying existing student result details.

```
Choice: 7
Roll No: 24B11CS383
Course Code: 241EE001
New Marks: 90
✅ Result Updated
```

Figure 8: Student Updation Screen of Student Result Management System

View Student Result Screen

Displays all Student result records in a table format.

```
Choice: 9
24B11CS508 | 241CS001 | 93
24B11CS508 | 241MA001 | 88
24B11CS508 | 241PH002 | 76
24B11CS383 | 241EE001 | 90
24B11CS508 | 241CS010 | 97
```

Figure 9: Student Information Screen of Student Result Management System

Student Interface

```
=========== STUDENT RESULT ===========
Roll No : 24B11CS508
Name    : Varun
Dept    : CSE
Sem     : 4
Section : 5

-------------------------------------------------------------------
No    Code         Course                    Marks  Grade  Credits
-------------------------------------------------------------------
1     241CS001     C Programming             93     O      10
2     241MA001     Linear Algebra & Calculus 88     A+     9
3     241PH002     Modern Physics            76     A      8
4     241CS010     ADSA                      97     O      10
-------------------------------------------------------------------
Total : 354
Avg   : 88.5
Result: PASS

Process finished with exit code 0
```

7. CONCLUSION

The Student Result Management System has been successfully developed to manage student information and academic results in an efficient and organized manner. The system replaces manual result management with a computerized solution, thereby reducing errors, saving time, and improving data accuracy.

The application provides essential functionalities such as adding, updating, deleting, and searching student records, along with managing subject-wise results through a simple console-based interface. By using technologies like Java, JDBC, and MySQL, the system ensures reliable performance, secure data handling, and easy access to student information.

Overall, the system improves the efficiency of managing student data and reduces the workload of administrative staff. It also provides a scalable solution that can be further enhanced in the future by adding features such as graphical user interface, web-based access, and advanced result analysis.

In conclusion, the Student Result Management System meets the project objectives and serves as an effective tool for managing student academic records in educational institutions.

8. FUTURE ENHANCEMENTS

The Student Result Management System developed in this project provides basic functionalities for managing student records and academic results. However, the system can be further enhanced by adding advanced features to improve its functionality and usability.

Some possible future enhancements include:

- **Result Analysis and CGPA Calculation:**
  Adding features to calculate CGPA, overall performance, and subject-wise analysis of students.
- **Attendance Management:**
  Integrating attendance tracking and linking it with student performance for better academic monitoring.
- **Web-Based Application:**
  Converting the console-based application into a web-based system so that students and administrators can access it from anywhere.
- **Multi-User Support:**
  Allowing multiple users (admin and students) to access the system simultaneously with proper authentication.
- **Improved User Interface:**
  Developing a graphical user interface (GUI) using JavaFX or Swing to make the system more interactive and user-friendly.
- **Report Generation:**
  Adding functionality to generate downloadable result reports in PDF format.

These enhancements can make the system more scalable, secure, and suitable for real-time organizational use.

9. REFERENCES

The following references were used during the development of the Student Result Management System:

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APPENDIX

A.1 Sample Code

Database Connection Code:

The following code establishes a connection between the Java application and the MySQL database using JDBC.

DBConnection.java:

```
import java.sql.Connection;
import java.sql.DriverManager;
public class DBConnection {
private static final String URL =
"jdbc:mysql://localhost:3306/student_result_db1";
    private static final String USER = "root";
    private static final String PASSWORD = "your_password";
    public static Connection getConnection() {
      try {
          return DriverManager.getConnection(URL, USER, PASSWORD);
      } catch (Exception e) {
          System.out.println("Database connection failed");
          return null;
      }
    }
}
```

Insert Student Record:

This code is used to add a new student into the database.

InsertStudent.java:

```
import java.sql.Connection;
import java.sql.PreparedStatement;
public class InsertStudent {
    public static void main(String[] args) {
        try {
            Connection con = DBConnection.getConnection();
            String query = "INSERT INTO students VALUES (?, ?, ?, ?, ?, ?)";
            PreparedStatement pst = con.prepareStatement(query);
```

```
            pst.setString(1, "24B11CS999");
            pst.setString(2, "Test Student");
            pst.setString(3, "CSE");
            pst.setInt(4, 4);
            pst.setString(5, "A");
            pst.setString(6, "test123");
            int i = pst.executeUpdate();
            if (i > 0)
                System.out.println("Student Added Successfully");
            con.close();
        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

A.2 SQL Queries

Create Students Table:

```
CREATE TABLE students (
        roll_no VARCHAR(20) PRIMARY KEY,
        name VARCHAR(100) NOT NULL,
        dept VARCHAR(50),
        sem INT,
        section VARCHAR(10),
        password VARCHAR(50)
);
```

Create Results Table:

```
CREATE TABLE results (
    id INT AUTO_INCREMENT PRIMARY KEY,
    roll_no VARCHAR(20),
    course_code VARCHAR(20),
    course_name VARCHAR(100),
    marks INT,
    grade VARCHAR(5),
    credits DOUBLE,
    FOREIGN KEY (roll_no) REFERENCES students(roll_no)
);
```